A sharp ReLU energy baseline and compression above that baseline

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Abstract

For bias-free ReLU layers following an augmented Boolean input, every nonconstant classifier with global unit margin has Gaussian weighted energy at least $2L + \log d - 2\log 2$. A stronger finite-depth bound is attained exactly by a dictator at $d = 1$. We also prove a function-level compression theorem. If a representation with H hidden layers has Frobenius product P and $\beta = \log(dP^2/2^H)$, the same classifier admits a ReLU representation of depth $O(1+\beta)$ and product $\exp(O(1+\beta))$, independently of its original depth and widths. The proof combines a simultaneous separation bound for all opposite-labelled hidden pairs with a constant-branching cover and an explicit nearest-prototype classifier. An Occam argument gives a cold-prior sample bound polynomial in the target's relative product, independently of learner width and depth. The relative product can be exponential in target depth, so this does not prove a joint polynomial-depth learning theorem or an exponential learning lower bound. No corresponding result for tanh is asserted.

1 Model and statements

Let $d \geq 1$ and $\bar{x} = (1, x) \in \mathbb{R}^{d+1}$ for $x \in \{-1, 1\}^d$. A network with $H \geq 1$ hidden layers is

$$h_0(x) = \bar{x}, \qquad h_s(x) = \mathrm{ReLU}(W_s h_{s-1}(x)) \ (1 \leq s \leq H), \qquad z(x) = v^T h_H(x).$$

All hidden widths are finite and unrestricted. There are no separately supplied biases after the augmented input. Set $L = H + 1$ and

$$P = \|v\|_2 \prod_{s=1}^{H} \|W_s\|_{\mathrm{F}}.$$

A Boolean function f has global unit margin when $f(x)z(x) \geq 1$ for every cube input. At common width N, the Gaussian weighted energy is

$$\kappa = \frac{d}{N}\|W_1\|_{\mathrm{F}}^2 + \sum_{s=2}^{H} \|W_s\|_{\mathrm{F}}^2 + N\|v\|_2^2.$$

The product of these L weighted layer energies is exactly dP^2. All logarithms are natural unless a base is displayed.

Theorem 1 (Baseline and sharp one-variable example)**.** *If f is nonconstant and globally unit-margined, then*

$$P^2 \geq (1 + d^{-1/H})^H, \qquad dP^2 \geq (1 + d^{1/H})^H \geq 2^H\sqrt{d}.$$

Consequently

$$\kappa \geq L(1 + d^{1/H})^{H/L} \geq L(2^H\sqrt{d})^{1/L} \geq 2L + \log d - 2\log 2.$$

For $d = 1$ and $f(x) = x$, the first energy bound is attained at every common width $N \geq 2$. In particular its exact value is $L2^{H/L}$.

Theorem 2 (Compression above the baseline)**.** *Suppose a nonconstant f has a globally unit-margined representation with H hidden layers and product P. Put*

$$\beta = \log \frac{dP^2}{2^H}, \qquad G = 1 + \beta.$$

There is a universal constant C such that f has another globally unit-margined ReLU representation with at most CG hidden layers and Frobenius product at most $\exp(CG)$. There is no dependence on the original widths, individual coefficients, precision, or robustness.

Corollary 3 (Energy form)**.** *If $\kappa \leq 2L + B$ with $B \geq 0$, then the conclusion of Theorem 2 holds with*

$$G \leq C(1 + B).$$

In particular the new depth is $O(B + 1)$ and its product is $\exp(O(B + 1))$, independently of the original depth. This is a statement about an existing low-energy representation; it is not a learning theorem.

2 Two-anchor linear replacement

The Boolean cube graph is connected, so a nonconstant f has a bichromatic edge. Orient it as x_+, x_- with labels $+1, -1$. Write

$$X = [\bar{x}_+, \bar{x}_-], \qquad a_s = h_s(x_+), \quad b_s = h_s(x_-), \quad G_s = \begin{pmatrix} \|a_s\|^2 & a_s^T b_s \\ a_s^T b_s & \|b_s\|^2 \end{pmatrix}.$$

The hidden anchors are nonnegative. Neither vanishes, and they cannot be positively proportional: every subsequent layer and the readout preserve positive proportionality, which would contradict their opposite output signs. Thus every G_s is positive definite with nonnegative off-diagonal. The augmented input Gram has eigenvalues $2d, 2$, so

$$G_0 = X^T X = \begin{pmatrix} d+1 & d-1 \\ d-1 & d+1 \end{pmatrix}, \qquad \det G_0 = 4d.$$

Its off-diagonal is also nonnegative.

Lemma 4 (Linear replacement with a squared-error bound)**.** *Let two linearly independent anchors have Gram $G = \left(\begin{smallmatrix} A & C \\ C & B \end{smallmatrix}\right)$ with $C \geq 0$. For any row w, there is a minimum-norm linear row b that maps these anchors to their two ReLU outputs, and*

$$\|w - b\|^2 \leq \|w\|^2 - \|b\|^2.$$

Consequently, if W maps the anchors through ReLU to output Gram G', its least-Frobenius linear replacement B_* satisfies

$$\|W - B_*\|_{\mathrm{F}}^2 \leq \|W\|_{\mathrm{F}}^2 - \|B_*\|_{\mathrm{F}}^2, \qquad \|B_*\|_{\mathrm{F}}^2 = \operatorname{tr}(G'G^{-1}).$$

The two nonzero squared singular values of B_* are the eigenvalues of $G^{-1/2}G'G^{-1/2}$.

Proof. If both outputs are positive, project w onto the anchor span. If both are zero, use $b = 0$. In the remaining case, interchange anchors if needed so that the outputs are $p > 0, 0$ and the original preactivations are p, t with $t \le 0$. The least-norm replacement has evaluations $(p, 0)$. The difference has evaluations $(0, t)$ plus an orthogonal component, hence

$$\langle b, w - b \rangle = -Cpt/\det G \ge 0.$$

Expansion of $\|w\|^2$ proves the claimed inequality. Apply this rowwise. The matrix and singular-value formulas follow from the minimum-norm map YX^+ between two full-rank anchor-column matrices X, Y. □

It follows that for every hidden layer, including the first,

$$\|W_s\|_{\mathrm{F}}^2 \ge \operatorname{tr}(G_s G_{s-1}^{-1}) \ge 2\sqrt{\det G_s/\det G_{s-1}}. \tag{1}$$

The least-norm readout with evaluations $y_+ \ge 1, y_- \le -1$ costs $y^T G_H^{-1} y$. Nonnegativity of the off-diagonal gives

$$\|v\|^2 \ge \frac{\|a_H\|^2 + \|b_H\|^2 + 2a_H^T b_H}{\det G_H} \ge \frac{2}{\sqrt{\det G_H}}. \tag{2}$$

Multiplying (1) and (2) already proves $P^2 \ge 2^H/\sqrt{d}$.

3 The stronger baseline and its sharpness

For completeness we prove the stronger product inequality in Theorem 1. Replace all hidden matrices by the maps $B_1, \dots, B_H$ from Lemma 4, and let $B = B_H \cdots B_1$ and $V = \|v\|$. The product has rank two, transports X exactly to $[a_H, b_H]$, and vanishes off the input anchor span. Let $\sigma_1 \ge \sigma_2 > 0$ be the two singular values of VB. The input-edge difference has norm two, while the output difference is at least two. Therefore $\sigma_1 \ge 1$. Also (2) gives

$$\sigma_1 \sigma_2 = V^2 \sqrt{\frac{\det G_H}{\det G_0}} \ge d^{-1/2}.$$

The rank-two form of Schatten Hölder gives

$$P \ge V \prod_s \|B_s\|_{\mathrm{F}} \ge \|VB\|_{S_{2/H}},$$

including when $2/H < 1$. Here is a direct proof of the inequality needed. Let $a_s \ge b_s > 0$ be the singular values of the restriction of B_s between its two anchor planes, and let $u_1 \ge u_2 > 0$ be those of B. Area multiplication gives $u_1 u_2 = \prod_s a_s b_s$, while operator-norm submultiplicativity gives

$$\tau := \log(u_1/u_2) \le \sum_s \log(a_s/b_s).$$

Put $t_s = \log(a_s/b_s) \ge 0$. Convexity of $\log\cosh$ and monotonicity of cosh on the positive half-line imply

$$\prod_s \|B_s\|_{\mathrm{F}}^2 = 2^H \Big(\prod_s a_s b_s\Big) \prod_s \cosh t_s \ge 2^H u_1 u_2 \cosh(\tau/H)^H = \big(u_1^{2/H} + u_2^{2/H}\big)^H.$$

Multiplication by V proves the claimed inequality for VB without requiring a separate quasi-norm theorem. Thus

$$P^{2/H} \ge \sigma_1^{2/H} + \sigma_2^{2/H} \ge \min_{t \ge 1} \{t^{2/H} + d^{-1/H} t^{-2/H}\} = 1 + d^{-1/H}.$$

The last function is nondecreasing on $[1,\infty)$. This proves the product bound. Weighted-layer AM–GM proves the first energy bound. Finally

$$L(2^H\sqrt{d})^{1/L} = 2L\exp\left(\frac{\frac{1}{2}\log d - \log 2}{L}\right) \geq 2L + \log d - 2\log 2.$$

For $d = 1$, use the first-layer indicator rails

$$\mathrm{ReLU}((1+x)/2), \qquad \mathrm{ReLU}((1-x)/2),$$

then the two-dimensional identity in every internal layer and readout $(1,-1)$. The first squared Frobenius norm is one, every internal squared norm is two, and the readout squared norm is two, giving $P^2 = 2^H$. Pad by zero neurons to any width $N \geq 2$. If $e_1,\dots,e_L$ are the weighted energies and $g = 2^{H/L}$, multiply each corresponding weight block by $\sqrt{g/e_s}$. Their product is one. Positive homogeneity preserves the output and makes every weighted energy equal to g, attaining Lg. This proves Theorem 1.

Remark 5. For margin $a > 0$, multiply the product-squared lower bounds by a^2 and the energy bounds by $a^{2/L}$. Constants have a different baseline: a single positive rail attains $\kappa = Ld^{1/L}$. Its optimality follows from $P \geq 1$ for any globally unit-margined function: the input second moment is the identity, and successive zero-preserving contractions imply $\mathbb{E}z(x)^2 \leq P^2$.

4 All opposite-labelled hidden pairs remain separated

The margin inequalities yield more than a bound for one selected edge. Fix *any* pair of opposite-labelled cube inputs and let G_s be its hidden Gram. Both states are nonzero, nonnegative, and nonproportional at every stage. Lemma 4 therefore applies throughout its suffix. With

$$S_j = \prod_{s\leq j}\|W_s\|_{\mathrm{F}}, \qquad T_j = \|v\|\prod_{s=j+1}^{H}\|W_s\|_{\mathrm{F}},$$

for $1 \leq j \leq H$ we have $S_jT_j = P$ and

$$T_j^2 \geq \frac{2}{\sqrt{\det G_H}}\prod_{s=j+1}^{H} 2\sqrt{\frac{\det G_s}{\det G_{s-1}}} = \frac{2^{H-j+1}}{\sqrt{\det G_j}}.$$

Consequently

$$\sqrt{\det G_j} \geq 2^{1-j}dS_j^2e^{-\beta}. \tag{3}$$

Each hidden state has norm at most RS_j, where $R = \sqrt{d+1}$. The parallelogram area of two vectors is at most the first vector's norm times their distance. Dividing (3) by RS_j proves the simultaneous normalized separation bound

$$\left\|\frac{h_j(x)}{S_j} - \frac{h_j(y)}{S_j}\right\| \geq \Delta_j := \frac{2d}{R}\,2^{-j}e^{-\beta} \qquad \text{whenever } f(x)\neq f(y). \tag{4}$$

There is no original-depth factor in this estimate. It bounds precisely those pairs that require opposite final signs, rather than an unrestricted suffix Lipschitz constant. The baseline also gives $\beta \geq \frac{1}{2}\log d \geq 0$.

5 A cover and an explicit ReLU classifier

Lemma 6 (A universal radius-quartering cover)**.** *For $K = 17^{64}$, every finite matrix W maps any radius-r ball into at most K balls of radius $r\|W\|_{\mathrm{F}}/4$. The same holds after ReLU.*

Proof. Normalize $\|W\|_{\mathrm{F}} = 1$. Retain its largest sixty-four singular directions; the tail has operator norm at most $1/8$. Cover the unit ball in the retained space by a $1/8$-net of size at most 17^{64}. Orthogonality of the top and tail errors bounds the total radius by $\sqrt{2}/8 < 1/4$. Translation and scaling handle arbitrary balls, and ReLU is one-Lipschitz. □

Set

$$j_0 = \left\lceil \log_2(64e^{\beta}) \right\rceil .$$

If $H < j_0$, the original network already has the depth and product required by Theorem 2, since $\log P = (H \log 2 + \beta - \log d)/2$. Assume $H \geq j = j_0$. Repeated application of Lemma 6 covers the normalized states h_j/S_j by at most K^j balls of radius $R4^{-j}$. The simultaneous separation (4) makes every such ball monochromatic. Replace each nonempty ball center by an actual prefix image and retain its label. There are $q \leq K^j$ prototypes, each of norm at most R, and the covering radius doubles. Since

$$\frac{2R4^{-j}}{\Delta_j} = \frac{R^2}{d} e^{\beta} 2^{-j} \leq \frac{1}{32},$$

every query has a correct-label prototype within $\Delta_j/4$, while every opposite-label prototype is at distance at least Δ_j. Write $\Delta = \Delta_j$ below. Let $\mathcal{P}_+, \mathcal{P}_-$ denote the two nonempty prototype classes. Define

$$s_a(u) = 2\langle a, u\rangle - \|a\|^2, \quad g = \Delta^2/2, \quad t_{ab}(u) = \frac{s_a(u) - s_b(u)}{g},$$

for $a \in \mathcal{P}_+$ and $b \in \mathcal{P}_-$. Compute

$$\begin{aligned} C_{ab}(u) &= \frac{\mathrm{ReLU}(t_{ab}(u) + 1) - \mathrm{ReLU}(t_{ab}(u) - 1)}{2}, \\ A_a(u) &= \mathrm{ReLU}\left(\sum_{b \in \mathcal{P}_-} C_{ab}(u) - |\mathcal{P}_-| + 1 \right), \\ z_*(u) &= 2 \sum_{a \in \mathcal{P}_+} A_a(u) - 1. \end{aligned}$$

These formulas use two ReLU hidden layers and one linear readout. Always $C_{ab}, A_a \in [0, 1]$. On a positive query a close positive prototype beats every negative score by at least $\Delta^2 - \Delta^2/16 > g$, so its A_a equals one and $z_* \geq 1$. On a negative query, one close negative prototype beats every positive score by the same amount; every A_a vanishes and $z_* = -1$. The comparison cancels the squared query norm, which is why the suffix cost does not depend on the original feature dimension.

Positive homogeneity computes the normalized prefix using $W_s/\|W_s\|_{\mathrm{F}}$. Carry a constant rail of value one through it, starting from the augmented input. Every augmented prefix matrix has Frobenius norm $\sqrt{2}$, so its product is $2^{j/2}$. The comparison, A, and readout norms are respectively at most

$$Cq(1 + R^2/g), \qquad Cq^{3/2}, \qquad C\sqrt{q},$$

including constant-rail rows. Since $R^2/g = R^4 4^j e^{2\beta}/(2d^2) \leq 2 \cdot 4^j e^{2\beta}$ and $q \leq K^j$, the new product is bounded by

$$2^{j/2} C q^3 (1 + R^2/g) \leq \exp(C(j + \beta + 1)) = \exp(CG).$$

Its hidden depth is $j + 2 = O(G)$. This proves Theorem 2.

Finally, if $\kappa \le 2L + B$, weighted-layer AM–GM gives

$$\frac{dP^2}{2^H} \le 2\left(1 + \frac{B}{2L}\right)^L \le 2e^{B/2}.$$

Thus $\beta \le \log 2 + B/2$, proving Corollary 3.

6 A strong-prior Occam consequence

We make the statistical consequence explicit. The result in this section concerns the cold prior limit at each fixed finite learner width. It is not a uniform polynomial-width theorem at a fixed exponent gap.

Lemma 7 (Finite classification capacity)**.** *For ReLU networks of hidden depth at most h, Frobenius product at most $Q \ge 1$, and global unit margin on the d-cube, the number of distinct Boolean functions is at most*

$$\exp(\mathrm{poly}(d, h, Q)),$$

for one universal polynomial, independently of the original widths. Consequently the functions with any-depth relative product $dP^2/2^H \le T$, together with constants, form a class $\mathcal{F}_T$ with $\log|\mathcal{F}_T| \le \mathrm{poly}(d, \max(1, T))$.

Proof. We give the finite-width step to clarify the absence of a width parameter. For consecutive matrices W, V, let a_i be the norm of row i of W, b_i the norm of column i of V, and $A = \sum_i a_i b_i \le \|W\|_{\mathrm{F}}\|V\|_{\mathrm{F}}$. Ignore zero terms. Sample m indices with probabilities $a_i b_i / A$. Use incoming row w_i/a_i and outgoing column $Av_i/(mb_i)$ for each sampled neuron. Positive homogeneity makes the new preactivation at the next layer an unbiased approximation to the original. Its two-layer Frobenius product is exactly A, for every draw.

If the earlier prefix product is S, each sampled vector has norm at most $AS\sqrt{d+1}$. The suffix is bounded by its Frobenius product. Hilbert-space bounded-difference concentration and a union bound over the 2^d inputs therefore give final-output error at most ε/h with $m = \mathrm{poly}(d, h, Q, \varepsilon^{-1})$ samples. Process consecutive pairs from the first hidden layer through the last, using the readout as the final outgoing matrix. The product never increases, all retained widths are polynomial, and the total error is at most ε. Take $\varepsilon = 1/4$ and double the final readout.

Positive homogeneous rescaling now makes every hidden matrix have Frobenius norm one, while the readout norm is at most $2Q$. All hidden states have norm at most $\sqrt{d+1}$. Round the polynomially many matrix entries on an inverse-polynomial mesh. The usual layerwise one-Lipschitz recurrence gives uniform output error less than $1/4$. There are only $\exp(\mathrm{poly}(d, h, Q))$ resulting finite parameter descriptions, and each original function is represented by one of them. Theorem 2 gives the second assertion. □

Fix a target f with a unit-margined depth-$D \ge 2$, width-C ReLU representation of product P_*, and put

$$T_* = \max\{1, dP_*^2/2^{D-1}\}.$$

The learner uses depth $L \ge D$ and common width $N \ge \max(C, 2)$. In weighted energy coordinates its raw prior has density proportional to $\exp(-\kappa/(2\sigma^2))$. Its posterior conditions on strict absolute margin $|z(x)| > 1/2$ at every cube input and agreement with the noiseless sample labels. The input distribution below is arbitrary.

Theorem 8 (Width- and learner-depth-independent cold sample bound)**.** *There is a universal polynomial p such that, for all $\alpha, \delta \in (0,1)$, the sample bound*

$$n \geq \frac{6}{\alpha}\left[p(d, T_*) + \log\frac{1}{\delta}\right]$$

has the following property. For each fixed admissible L, N, and all sufficiently small positive σ (with a threshold allowed to depend on L, N, α and the target), with sample probability at least $1-\delta$ the posterior assigns the correct label probability strictly greater than $2/3$ on an input-distribution fraction at least $1-\alpha$. The displayed sample bound is independent of the learner width and depth. Its dependence on the target product may be exponential in target depth.

Proof. First produce an exact target witness at the learner architecture. When $L > D$, replace the scalar readout by the two rails $\mathrm{ReLU}(z), \mathrm{ReLU}(-z)$ and final readout $(1, -1)$, and insert two-rail identity layers as needed. The initial split multiplies the product by two and adds one weighted layer. Every subsequent identity layer multiplies its square by two. Thus the padded target product P_0 satisfies

$$T_0 := dP_0^2/2^{L-1} \leq 2T_*.$$

For $L = D$ use the target directly. Pad hidden layers to a common base width $C_0 \leq \max(C, 2)$. At any $N \geq C_0$, clone every base neuron $k = \lfloor N/C_0 \rfloor$ times and pad the remaining coordinates by zero. The first Frobenius norm gains $\sqrt{k}$, internal norms are unchanged, and the readout norm gains $1/\sqrt{k}$. Hence the function and its product are preserved exactly, including at widths that are not multiples of C_0.

Balance the L weighted layer energies by positive homogeneous scaling. The exact target witness then has energy

$$K = L(dP_0^2)^{1/L} = Lg, \qquad g \geq 1.$$

Here $g \geq 1$ follows from $P_0 \geq 1$, using the input second moment and global margin. The witness has strict conditioning margin. A fixed neighborhood therefore lies in every sample-conditioning event and has energy at most $K + 1/4$. Finite-dimensional Gaussian Laplace comparison shows, uniformly over all samples,

$$Q_{S,\sigma}\{\kappa > K + 1\} \longrightarrow 0 \quad (\sigma \downarrow 0).$$

The denominator can use the same target neighborhood for every sample. No limit uniform in L, N is asserted.

For a posterior network of energy at most $K + 1$, AM–GM gives

$$dP^2 \leq ((K+1)/L)^L \leq e^{1/g} dP_0^2 \leq e\, dP_0^2.$$

Its margin is greater than $1/2$. Doubling its readout gives a unit-margin representation with relative product at most $4eT_0 \leq 8eT_*$. Constant classifiers are harmless separately. Lemma 7 therefore puts all these posterior functions in one class $\mathcal{F}$ with $\log|\mathcal{F}| \leq p(d, T_*)$, independently of L, N.

A union bound over $\mathcal{F}$ shows that with sample probability at least $1-\delta$, every sample-consistent member has population error at most $\alpha/6$. Choose the cold threshold so the posterior mass outside this class is at most $\alpha/12$. Its Gibbs error is then at most $\alpha/4$. If the posterior correct-label probability is at most $2/3$ at an input, its error probability there is at least $1/3$. Markov's inequality bounds the fraction of such inputs by $3\alpha/4 < \alpha$, proving the stated strict-confidence conclusion. □

7 Scope and further consequences

The structural theorem concerns the represented Boolean function, rather than the stability of a particular parameterization. It permits arbitrary original widths and arbitrarily small hidden amplitudes. The product-to-parallel-circuit theorem in the companion paper [1] also gives threshold circuits of size $\exp(O(\beta+1))$ and depth $O(\beta+1)$, and Boolean circuits of the same size with depth $O((\beta+1)^2)$. Its proof is separate from the arguments above.

The cold sample theorem does not imply a polynomial sample bound in arbitrary target depth: T_* itself can be exponential in that depth. An extension at prescribed finite Gaussian scales requires a separate posterior comparison; it is not proved here. The cold threshold in Theorem 8 is allowed to depend on the fixed architecture.

No natural-activation exponential sample lower bound is proved here. No assertion about the original unrestricted tanh learning problem follows from these ReLU results.

Verification scope. The arguments have received independent conventional mathematical audits. The complete results in this paper have not been formalized in Lean.

References

[1] ChatGPT-6 Astra, Kaarel Hänni, and Dmitry Vaintrob. *Frobenius product, parallel circuits, and sample-complexity barriers for deep Gaussian Bayesian networks.* Research draft, 25 October 2026. https://github.com/kaarelh/AI-works/tree/main/neural-network-generalization/works-in-progress/boolean-depth-barriers.